

also be done using third party software and cloud systems as per end user requirements.

- NAS devices come with fully functional operating systems, including access to additional applications. This enables end users to download a variety of applications to extend the NAS device functionality including live IP camera monitoring, backup-cum-disaster recovery, etc.
- NAS technology based appliances are small and compact in size, so they consume less power and are highly portable as compared to other servers.

Top open source software for NAS

Backup and recovery software play a critical role for small, medium and even large enterprises, and ensure data is not lost in case of any natural or technical disaster. Home users, too, require backup software to have a lifetime record of all their memories and documents. NAS is a one stop solution for efficient backup of data. Here are the top open source software solutions for NAS that are efficient and easy to use.

FreeNAS

FreeNAS is open source-cum-free network attached storage software based on FreeBSD and the OpenZFS file system. It is licensed under the BSD licence and runs on X86-64 hardware. It provides a simple and comprehensive GUI to systems administrators to create a centralised and easily accessible place for data access. The GUI interface supports all standard administration features like storage pool configuration, user management, sharing options and maintenance. FreeNAS provides strong support for data integrity, anti-data corruption, time snapshotting, replication and lots of levels of redundancy like striping, mirroring, and all three levels of RAID through the OpenZFS file systems.

Features

- Almost every file sharing is supported via FreeNAS, which includes major file systems like SMB/CIFS (Windows file shares), NFS (UNIX files), AFP (Apple file shares), FTP, iSCSI, and WebDAV. It also supports VMware VAAI, Microsoft ODX and Server 2008/2012 R2 clustering.

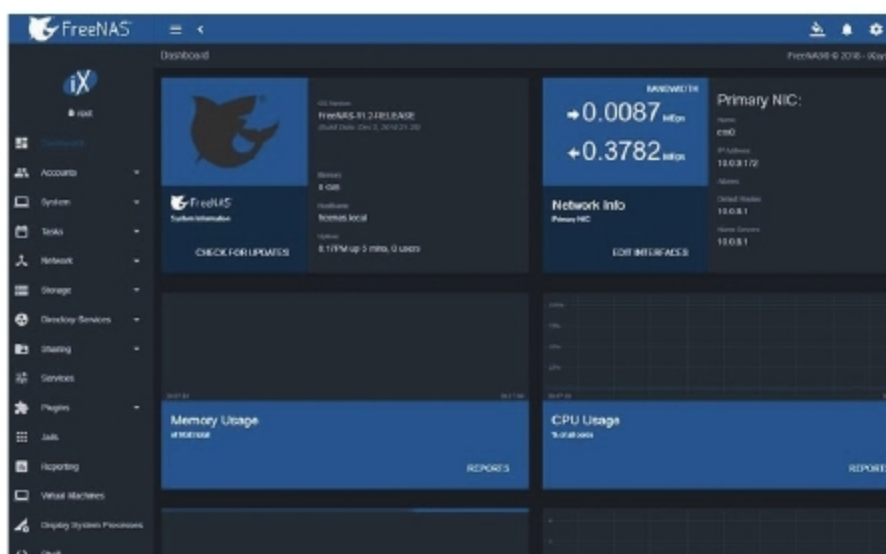


Figure 3: FreeNAS GUI

- It has an angular based Web UI for simplifying complex admin tasks like volume creation, security based permissions, software updates and even disaster recovery.
- A highly resilient ZFS file system provides high end compression, full volume encryption, snapshots, data duplication, and user quotas.
- It has numerous features with regard to network services like support for standard network protocols, LDAP (lightweight directory access protocol)/active directory, VLAN networking, dynamic DNS client, SNMP monitoring and for a wide range of network hardware.
- It supports lots of plugins like Bacula, CouchPotato, BitTorrent, ownCloud, etc.

Latest version: 11.2-RC2

Official website: <https://www.freenas.org/>

XigmaNAS

XigmaNAS is open source NAS server software with a user friendly interface designed in PHP language. It is primarily designed from FreeNAS 7 code and well updated with all the latest FreeBSD releases. XigmaNAS supports sharing across Windows, Apple and UNIX-like systems. It includes ZFS v5000, software RAID (0,1,5), disk encryption, Self-Monitoring, Analysis, and Reporting Technology (S.M.A.R.T.) and email reports, etc, with the following protocols: SMB, Samba Active Directory Domain Controller AD, FTP, NFS, TFTP, AFP, rsync, Unison (file synchroniser), iSCSI (initiator and target), HAST, CARP, Bridge, UPnP and BitTorrent.

Features

- **Web based interface:** XigmaNAS is powered by a fast, stable and responsive Web GUI, which allows the entire server to be configured and managed without the need to interact with the FreeBSD command line.
- **Data sharing:** It supports all operating systems like Linux, MAC, Windows, iOS, Android with tons of protocols like SMB, LDAP, active directory, FTP, SFTP, NFS, SSH, UPnP, BitTorrent and Bridge, CARP and HAST.
- **Boot options:** XigmaNAS can be booted from a LiveCD or LiveUSB, and can even be installed on compact Flash, USB flash drive, SSD, hard drive or a number of other bootable devices.

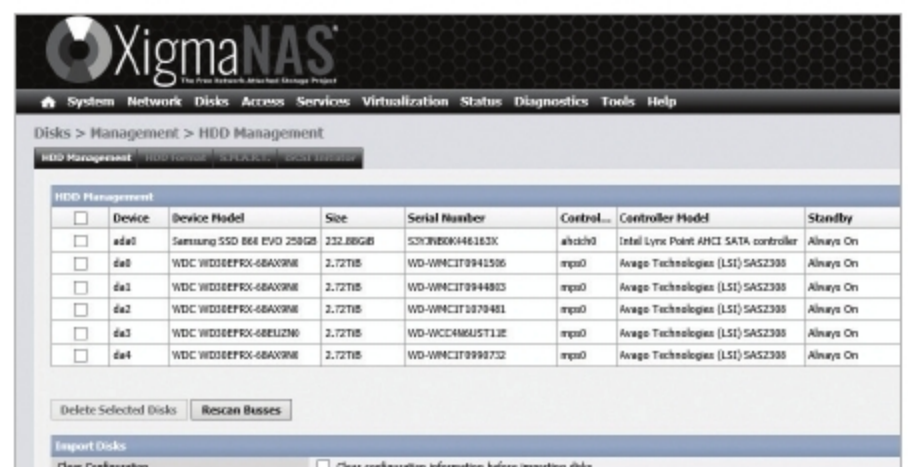


Figure 4: XigmaNAS